

NEURONFORGE TECHNOLOGIES

Product & Innovation Team

TECHNICAL SPECIFICATION

PRD Genie

An Agentic AI Assistant for Product Managers, built on n8n

Capstone Project 2 — Applied Agentic AI for PMs
Interview Kickstart

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Problem Statement • Deliverables • Milestones

Document Control

Field	Detail
Document Title	PRD Genie — Agentic AI PM Assistant: Technical Specification
Project	Capstone Project 2 — Applied Agentic AI for PMs
Author	Animesh
Version	1.0
Status	Draft for final review
Build Platform	n8n (self-hosted / cloud instance)

1. Introduction

1.1 Background

NeuronForge Technologies is a fast-growing AI product company. Its Product Managers spend a disproportionate share of their time scoping calls and stakeholder meetings, then manually convert those discussions into Product Requirement Documents (PRDs), epics, features, and user stories. This is slow, repetitive, and produces inconsistent documentation across teams.

1.2 Problem Statement

- PMs spend excessive time converting meeting transcripts and notes into structured requirements.
- PRD creation is manual, iterative, and not standardized across teams.
- Delays in documentation slow down epic/feature breakdown and time-to-market.

1.3 Objective of this Document

This specification defines the end-to-end design of PRD Genie — an agentic AI assistant that ingests meeting transcripts and product documents, and generates structured PRDs (with epics, features, and user stories) with a human-in-the-loop review step. The focus of this document is the n8n implementation: workflow structure, nodes, data contracts, and guardrails — kept deliberately simple and buildable within the capstone timeline.

2. Scope

2.1 In Scope

- Ingest meeting transcripts and supporting notes/documents (text-based) from **Slack Channel**.
- Extract and classify requirement statements from raw input.
- Generate a structured PRD: Overview → Epics → Features → User Stories (with acceptance criteria).
- Human-in-the-loop review/edit step before the PRD is finalized.
- Basic traceability — each generated item links back to the source snippet it came from using Chunks.

- Success-metric tracking (baseline vs. after-pilot).

2.2 Out of Scope (Non-Goals)

- Voice/audio transcription (assume transcripts are already texted — e.g., exported from Zoom/Teams).
- Direct write-back into JIRA/Confluence (a mocked/optional integration only, not core to MVP).
- Multi-language support — English only for the pilot.
- Fine-tuning or hosting a custom LLM — the pilot uses a hosted LLM via API.

2.3 Assumptions

- Input transcripts are plain text (.txt) or pasted text — cleaned of major transcription noise.
- Dummy/mock data will be used for meeting transcripts, product notes, and a sample backlog (documented in Section 7).
- One n8n workflow serves one "PRD request" per execution; concurrent requests are out of scope for the pilot.
- A single reviewer (the PM) performs the human-in-the-loop approval; no multi-approver routing in v1.

3. Solution Overview

PRD Genie follows a simple perceive → plan → act → review loop, mapped directly to n8n nodes:

Stage	What Happens	Automate or Augment
Perceive	Ingest transcript/notes; clean and chunk text.	Automate
Plan	Classify content into requirement types; extract key entities (actors, goals, constraints).	Automate
Act	Generate structured PRD draft — epics, features, user stories with acceptance criteria.	Automate
Review	PM reviews, edits, approves or sends back for regeneration via Slack Channel	Human-in-the-loop
Learn	Approved edits are logged to refine prompt examples over time.	Augment (manual, v1)

Design Principle

- Keep every AI step grounded in the source transcript — no fact should appear in the PRD that cannot be traced to an input snippet.
- Nothing is auto-published. The PM always approves before a PRD is considered final.
- Favor a small number of well-tested n8n nodes over a large, hard-to-maintain graph.

4. n8n Workflow Architecture

The workflow is a single n8n pipeline made of five stages. Each stage is a small group of nodes so the flow stays easy to debug and extend.

4.1 High-Level Flow

1. Trigger — Webhook / Form node receives a new PRD request (transcripts) from Slack.

2. Ingestion & pre-processing — clean, chunk, and normalize the input text.
3. Extraction & Classification — AI Agent node extracts requirement statements and classifies them.
4. PRD Generation — AI Agent node drafts the PRD (epics → features → user stories) using a Structured Output Parser.
5. Human-in-the-loop Review — draft is sent to the PM (Slack for approval or modification request with instructions to add/amend or remove any items in the PRD).
6. Draft PRD is saved in the N8N Database with Status as Pending or Modification Requested until Approved
7. Output & Logging — approved PRD is stored (N8N Database) and Status is updated.
8. Approved PRD is converted to PDF file and emailed to PM via Gmail

Text Flow Diagram

[Webhook/Form Trigger -Slack] → [Ingestion & Chunking] → [Extraction Agent] → [Structured Output Parser] → [PRD Generation Agent] → [Review via Slack (Human-in-the-loop)] → [Switch: Approved / Modify] → [Approved: Save to Doc + Log Metrics] / [Modify: Loop back to Generation with feedback] → [Approved PRD is converted to PDF and emailed to PM via Gmail]

4.2 Node-by-Node Design

#	n8n Node	Type	Purpose
1	New PRD Request	Webhook / Form Trigger-Slack	Accepts transcript text, meeting title
2	Clean & Chunk Text	Code / Function node	Strips filler words/timestamps; splits long transcripts into manageable chunks (e.g., ~2–3k tokens) for reliable LLM processing.
3	Extraction Agent	AI Agent node (Google Gemini)	Reads each chunk; extracts candidate requirement statements, actors, goals, and constraints as structured JSON.
4	Merge & Deduplicate	Code / Merge node	Combines extraction results across chunks; remove duplicate or near-duplicate requirement statements.
5	Classification & Prioritization	AI Agent node + Structured Output Parser	Tags each requirement as Epic / Feature / User Story; assigns a draft priority (High/Med/Low) with a one-line rationale.
6	PRD Generation Agent	AI Agent node (Open AI) + Structured Output Parser	Synthesizes the classified items into a structured PRD: Overview, Epics, Features, User Stories, Acceptance Criteria.
7	Traceability Tagger	Code node	Attaches the source chunk/snippet reference to each generated item for auditability.
8	Send for Review	Slack node + Approve/Modify request	Notifies the PM with Draft PRD with Epics, Features, User Stories and other information
9	Review Outcome	Switch node	Routes based on PM decision: Approve / Modify
10	Save Final PRD	N8N Database with Status	Writes the approved PRD into a Doc (narrative) and a Sheet (structured backlog: Epic/Feature/Story rows).

11	Email to PM	PDF/Gmail	Approved PRD is converted to PDF and emailed to PM via gmail.
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Why these node choices

- AI Agent nodes act as the reasoning/routing layer — consistent with prior workflow patterns (intent routing via Switch, Structured Output Parser for reliable JSON).
- Structured Output Parser is used at every LLM step that feeds a downstream node, so outputs are predictable JSON rather than free text.
- Simple Memory (session-scoped) is attached to the PRD Generation Agent only, so it can reference earlier chunks of the same transcript without carrying context across unrelated requests.

4.3 Data Contracts (Key JSON Shapes)

Keeping a fixed JSON shape between nodes is what makes the workflow debuggable. Two contracts matter most:

Extraction output (per chunk)

Field	Type	Notes
requirement text	string	Verbatim or lightly cleaned statement from the transcript.
actor	string	Who raised it (role, e.g. "Sales stakeholder").
type guess	enum	epic / feature / user_story / unclear.
source_ref	string	Chunk ID + line offset, for traceability.

PRD output (final)

Field	Type	Notes
overview	string	1–2 paragraph product summary generated from all chunks.
Success Metrics	array	Performance and usability of the product
epics	array	Each epic has: title, description, linked features.
features	array	Each feature has: title, description, linked user stories [], Source reference.
user_stories	array	"As a <actor>, I want <goal>, so that <benefit>" + acceptance criteria [].
Acceptance Criteria	array	To accept the product for the purpose
Constraints	array	Any conditions to be followed
open questions	array	Ambiguous points the agent could not resolve — surfaced to the PM, not guessed.

5. Data & Mock Integrations

Realistic dummy data will be created and documented, since no live NeuronForge systems are available for the pilot.

Data / Integration	Mock Approach
Meeting transcripts	5–10 sample transcripts (mix of scoping calls, stakeholder syncs), written as realistic dialogue with actors and requirement chatter.

Data / Integration	Mock Approach
Existing backlog / product notes	A small sample backlog (CSV or Sheet) with prior epics/features, used only as style reference.
JIRA / Confluence	Not connected live; a Google Sheet stands in for the backlog, structured to mirror a JIRA export.
Notification channel	Email (Gmail node) or Slack — whichever is already connected in the n8n instance.

All assumptions and mock data are documented alongside the workflow export, so the design choices are auditable during evaluation.

6. Guardrails

6.1 Grounding

- Every generated epic/feature/story must carry a `source_ref` back to the transcript chunk it came from.
- The PRD Generation Agent's system prompt explicitly instructs: "Do not invent requirements do not present in the source text; add unclear items to open questions instead."

6.2 Privacy

- Transcripts may contain names/roles — mask direct customer PII (emails, phone numbers) before sending to the LLM, using a simple regex step in the Clean & Chunk node.
- Only the minimum necessary text is sent per chunk; full transcripts do not persist beyond the run log.

6.3 Human-in-the-Loop

- No PRD is saved as final without explicit PM approval through the Review Outcome step.
- Modification requested drafts are looped back with the PM's comment appended to the prompt for regeneration.

7. Success Metrics & Baseline

Metric	Baseline (Week 0)	Target Direction
PRD drafting cycle time	Manual time to draft a PRD from a transcript (measured across 3–5 sample cases)	Reduce by ≥50%
Requirement extraction accuracy	% of manually identified requirements also caught by the agent	≥85% recall on sample set

PM edit effort on draft	Edit distance / % of draft rewritten before approval	Decreasing trend across runs
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Metric	Baseline (Week 0)	Target Direction
Approval rate on first pass	% of drafts approved without a full regeneration	Increasing trend across runs
Traceability coverage	% of generated items with a valid source_ref	100%

Baselining approach: before the pilot, manually draft PRDs for 3–5 sample transcripts and time for the process. Compare PRD Genie's output on the same transcripts for cycle time, accuracy, and edit effort.

8. Deliverables Checklist

- ☐ Working n8n workflow demonstrating ingestion → extraction → PRD generation → human review → save.
- ☐ n8n workflow export (JSON).
- ☐ Slide deck (≤9 slides) summarizing problem, approach, and outcomes.
- ☐ ~5-minute demo video walking through the live workflow.
- ☐ Sample/mock data set (transcripts, backlog reference) with assumptions documented.
- ☐ Success-metrics baseline and post-pilot comparison.

